Prodigy Anesthesia Quick Review

Autonomic Hyperreflexia/Dysautonomia

Why does autonomic hyperreflexia occur?	Stimulation of the anterolateral column of the spinal cord below the level of the lesion results in an increase in sympathetic activity. Normally, these impulses are counteracted by inhibitory impulses from higher centers of the nervous system. Because of the spinal cord injury, however, the inhibitory impulses are blocked, resulting in unimpeded sympathetic outflow. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 259.*
In what patient population does autonomic hyperreflexia occur?	It occurs in patients who have experienced spinal shock and appears in conjunction with the return of spinal reflexes. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 259.*
What are the hallmark cardiovascular symptoms of autonomic hyperreflexia?	Severe hypertension and reflex bradycardia. It may also be accompanied by blurred vision, loss of consciousness, seizures, cardiac dysrhythmias, and pulmonary edema. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 259.*
What factors can trigger autonomic hyperreflexia?	It can be triggered by cutaneous stimulation such as surgical incision or visceral stimulation such as distention of the bladder or rectum. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 259.*
What are methods of preventing autonomic hyperreflexia?	The goal is to prevent the triggering event from initiating the reflex by deep general anesthesia, neuraxial anesthesia, or regional anesthesia. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 259-260.*
How is the incidence of autonomic hyperreflexia related to the level of the spinal cord injury?	About 85% of patients with an injury above T6 will suffer from autonomic hyperreflexia, but it is unlikely to occur at all in patients with a lesion below T10. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 259.*
During anesthesia, when is autonomic hyperreflexia most likely to occur?	With the initiation of surgical stimulation at the beginning of the case, especially if the patient is not under a deep enough anesthetic, and at the end of the case as the effects of the anesthetic are wearing off. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 260.*
What is the treatment for autonomic hyperreflexia if it occurs?	Vasodilators with a short half-life (such as sodium nitroprusside) should be available to treat the sudden onset of systemic hypertension. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 260.*

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Will an epidural prevent autonomic hyperreflexia from occurring in a laboring parturient with a T6 injury?	Epidural anesthesia has been reported to be effective in preventing autonomic hyperreflexia from occurring due to uterine contractions, but, because epidurals may spare the sacral segments, spinal anesthesia is more protective. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 260.*
What symptoms may be seen in a patient who is awake and experiences autonomic hyperreflexia?	Severe headache and blurred vision may occur due to severe hypertension and nasal stuffiness may occur due to reflexive cutaneous vasodilation. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 260.*
What is reflexive cutaneous vasodilation and how does it relate to autonomic hyperreflexia?	When a patient experiences autonomic hyperreflexia, a reflex dilation of the tissues ABOVE the level of the spinal cord injury will occur to help offset the sudden increase in blood pressure. Patients may experience nasal stuffiness as a result of this effect. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 259-260.*

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